



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Continuity in Metric Space

Continuity, Uniform continuity, Homeomorphisms

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CONDUCTED BY

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Continuity of Functions in Metric Spaces

Continuity at a Point in $\mathbb{R}$

Let $f : A \rightarrow \mathbb{R}$ be a function defined on a subset A of $\mathbb{R}$, and let $c \in A$. We say that f is **continuous at the point** a if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\forall x \in A \text{ and } |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

Continuity at a Point in Metric Space

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a function. We say that f is **continuous at the point** $a \in X$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\forall x \in X \text{ and } d_X(x, a) < \delta \implies d_Y(f(x), f(a)) < \varepsilon.$$

Equivalent Definition using Open Spheres

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a function. We say that f is **continuous at the point** $a \in X$ if for every open sphere $S_\varepsilon(f(a))$ in Y , there exists an open sphere $S_\delta(a)$ in X such that

$$f(S_\delta(a)) \subseteq S_\varepsilon(f(a)).$$

Equivalent Definition using Open Set

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a function. We say that f is **continuous at the point** $a \in X$ if for every open set U in Y containing $f(a)$, there exists an open set V in X containing a such that

$$f(V) \subseteq U.$$

Continuous Functions

Continuity using Open Sets

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a function. We say that f is continuous at all points of X i.e. **continuous** if for any open set U in Y , the preimage $f^{-1}(U)$ is an open set in X .

Continuity using Closed Sets

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a function. We say that f is continuous at all points of X i.e. **continuous** if for any closed set C in Y , the preimage $f^{-1}(C)$ is a closed set in X .

Examples of Continuous Functions

💡 Continuity in Discrete Metric Space

Let (X, d_X) be a discrete metric space, and let (Y, d_Y) be any metric space. Then every function $f : X \rightarrow Y$ is continuous at every point of X .

💡 Continuity of a constant function

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a constant function defined by $f(x) = c$ for all $x \in X$, where $c \in Y$ is a fixed point. Then f is continuous at every point of X .

💡 Identity Function is Continuous

Let (X, d) be a metric space. Then the identity function

$$I_X : X \rightarrow X, \quad I_X(x) = x$$

is continuous at every point of X .



💡 Composition of Continuous Functions

Let (X, d_X) , (Y, d_Y) , and (Z, d_Z) be metric spaces. Suppose

$$f : X \rightarrow Y \quad \text{and} \quad g : Y \rightarrow Z$$

are continuous. Then the composition

$$g \circ f : X \rightarrow Z$$

is continuous.



💡 Lipschitz Functions are Continuous

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a function. Suppose there exists a constant $L > 0$ such that

$$d_Y(f(x), f(y)) \leq L d_X(x, y) \quad \text{for all } x, y \in X.$$

Then f is continuous on X .

Distance Function is Continuous

Let (X, d) be a metric space and fix a point $a \in X$. Define

$$f(x) = d(x, a), \quad x \in X.$$

Then $f : X \rightarrow \mathbb{R}$ is continuous.



💡 Norm is Continuous

The function

$$\| \cdot \| : \mathbb{R}^n \rightarrow \mathbb{R}, \quad x \mapsto \|x\|$$

is continuous.



Continuity in $\mathbb{R}^2$

💡 Continuity in $\mathbb{R}^2$

Let $f : A \rightarrow \mathbb{R}^2$ be a function defined on a subset A of $\mathbb{R}^2$, and let

$$f(x, y) = (f_1(x, y), f_2(x, y))$$

where f_1 and f_2 are real-valued functions. Then f is continuous at the point $(a, b) \in A$ if and only if both f_1 and f_2 are continuous at the point (a, b) .

Continuity in Complex Plane

Continuity in Complex Plane

Let $f : A \rightarrow \mathbb{C}$ be a function defined on a subset A of $\mathbb{C}$, and let $z_0 = a + ib \in A$. We say that f is continuous at the point z_0 if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\forall z \in A \text{ and } |z - z_0| < \delta \implies |f(z) - f(z_0)| < \varepsilon.$$

Continuity of Complex Functions in $\mathbb{R}^2$

Let $f : A \rightarrow \mathbb{C}$ be a function defined on a subset A of $\mathbb{C}$, and let

$$f(z) = u(x, y) + iv(x, y)$$

where $z = x + iy$ and u and v are real-valued functions. Then f is continuous at the point $z_0 = a + ib \in A$ if and only if both u and v are continuous at the point (a, b) .

Continuity using Sequences

💡 Continuity at a Point using Sequences

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a function. We say that f is **continuous at the point** $a \in X$ if and only if for every sequence (x_n) in X that converges to a , the sequence $(f(x_n))$ in Y converges to $f(a)$.

Uniform Continuity

Uniform Continuity

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a function. We say that f is **uniformly continuous** if for every $\varepsilon > 0$, there exists a $\delta > 0$, that depends only on ε , such that

$$\forall x, y \in X \text{ and } d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

💡 Cauchy sequence with uniform continuity

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a uniformly continuous function. Then for every Cauchy sequence (x_n) in X , the sequence $(f(x_n))$ in Y is also a Cauchy sequence.

② Not uniform continuity with Cauchy sequence

Give an example of a function $f : X \rightarrow Y$ that is continuous but not uniformly continuous, and a Cauchy sequence (x_n) in X such that the sequence $(f(x_n))$ is not a Cauchy sequence in Y .

Homeomorphisms

Homeomorphism

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \rightarrow Y$ is called a **homeomorphism** if it is a bijection and both f and its inverse $f^{-1} : Y \rightarrow X$ are continuous.

Homeomorphic Spaces

Let (X, d_X) and (Y, d_Y) be metric spaces. We say that X and Y are **homeomorphic** if there exists a homeomorphism $f : X \rightarrow Y$. In this case, we write $X \cong Y$.

Thank You!

We'd love your questions and feedback.

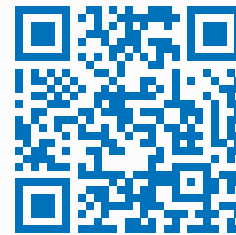
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(Lectures, walkthroughs, and course updates)



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References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.